

# Herald



## Herald — Continuation

| Field          | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| Revision       | 20                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Created        | 2026-05-20                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Last modified  | 2026-05-28                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Status         | active                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Status summary | <p><b>2026-05-28 (day-3) — Wave 8 Track B COMPLETE + v0.6.0 tagged (HEAD b313a5e; constitution at 15cd4bc; all 4 mirrors synced).</b> Wave 8 Track B (MTPProto user-account harness for §11.4.98 full-automation testing) is now end-to-end live-evidence-proven: HRD-139 qaherald/internal/mtproto/Client live wiring (238448e — gotd/td v0.144.0 vendored as a submodule, Connect/SendMessage/WaitForReply/WhoAmI/Close interface, 4198-byte encrypted session file at ~/.config/herald/mtproto.session, mode 0600)+ qaherald mtproto login/whoami/logout Cobra subcommands (ec4c3c0 — added HERALD_MTPROTO_ALLOW_NON_TTY=1 env override for delegated-by-conductor login per the operator "can you run it alone?" mandate; conductor-driven FIFO-piped OOB-code flow proven autonomous end-to-end with operator's single 5-digit chat-paste). HRD-140 TestMTPProto_Subscribe_AutonomousRoundTrip PASSED LIVE in 1.30s (367d58b — MTPProto sends a stimulus + polls bot getUpdates for delivery confirmation, replacing the legacy TestSubscribe_LiveBotAPI Bot-API-only path which is fundamentally non-autonomous per the §11.4.98 contract). HRD-141 TestMTPProto_Wave6_AutonomousClosedLoop PASSED LIVE in 48.58s (7e50ed0 then re-fixed in d7d6524 + 0797e85 — spawns pherald listen with cmd.Dir = repoRoot (CRITICAL — pherald resolves docs/Issues.md relative to its CWD), pre-acks pending updates via consumePendingUpdates(t, botToken) to prevent TLS-connection-reset storms from stale chat messages, asserts on the pherald --qa-out-dir JSONL journal (tgram.message in + cc.dispatch + cc.reply directions, matched on the unique test-text string rather than message_id — MTPProto's int64 message_id is a different namespace from Bot API's chat-local message_id) — replaces tests/test_wave6_live_loop.sh). HRD-142 TestMTPProto_Wave65_LifecycleAutonomous PASSED LIVE in 26.04s with 3/3 subtests PASS (1f2e75d — 3 representative Wave 6.5 fast-path scenarios (Help / Status / Continue), each assertion is tgram.message in + `cc.reply</p> |

| Field          | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| Issues         | HRD-008, HRD-015, HRD-019..HRD-025, HRD-029..HRD-056, HRD-081, HRD-085..HRD-090, HRD-115..HRD-121 (Wave 7 T6-T12 pending — Slack live evidence gated), HRD-131 (SQLite SSoT Phase 1, OPEN-deferred) — <b>31 open</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Issues summary | see <code>Issues.md</code> . r19 (Tgram-completeness wave COMPLETE) atomically migrated HRD-133/134/135/136/137/138 Issues→Fixed per §11.4.19 — see <code>Issues.md</code> r35 + <code>Fixed.md</code> r16. r18: v1.0.0 Batch A migrated HRD-018 / HRD-026 / HRD-027 Issues→Fixed (§11.4.19 atomic). v1.0.0 Batch B (HRD-019..025, §42 flavor constitution bindings) is <b>in_progress</b> — each flavor's <code>internal/bindings/</code> Pipeline + rule catalogue + PURE detectors + emit-through-audit foundation + §107.x evidence + §11.4.85 stress all landed (f33b5e4/5ebad4f/46ca77e/f73195f/3fb753a), but live operation-interception (the Subjects the detectors classify) is deferred to Batch C §43 command bodies, so these stay open not Fixed. Wave 7 closed HRD-110/111/112/113/114; HRD-115..121 are the remaining Wave 7 tasks T6-T12. HRD-122..HRD-130 (§11.4.85 stress+chaos suite, GAP-3) + HRD-132 (Runner idempotency-window bug) MIGRATED to <code>Fixed.md</code> . HRD-131 is §11.4.93 SQLite workable-items SSoT migration Phase 1 — <b>OPEN-deferred</b> per operator 2026-05-27: Phase 2 adoption blocked on the unimplemented + schema-incompatible constitution migration binary (assessment 36fa90b); a §11.4.74 constitution-repo task, not a Herald reimplementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Fixed          | HRD-001..HRD-018 (less open), HRD-026, HRD-027, HRD-028, HRD-080, HRD-092..HRD-101, HRD-110..HRD-114, HRD-122..HRD-130, HRD-132, HRD-133..HRD-138                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Fixed summary  | see <code>Fixed.md</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Continuation   | <p><b>Resume — live-evidence collection for tgram (HEAD 77da6eb):</b> The tgram-completeness wave is COMPLETE in code + tests + docs (HRD-133..138 Fixed). Hermetic coverage is full (tgram suite 26/26 PASS, 0 SKIP; chaos test deterministic). What remains is the LIVE-EVIDENCE layer — currently SKIP-with-reason in <code>scripts/e2e_bluff_hunt.sh</code> for E17/E18 (live Telegram send + roundtrip), E34 (live Claude Code session resolution), E63-E70 (Wave 6 closed-loop with real Telegram + Claude Code), E71-E80 (Wave 6.5 ticket lifecycle wire-byte). <b>Operator-action required:</b> export real credentials in the conductor's session via <code>! export HERALD_TGRAM_BOT_TOKEN=&lt;token&gt; HERALD_TGRAM_CHAT_ID=&lt;chat-id&gt; HERALD_TGRAM_BOT_USERNAME=&lt;username&gt; HERALD_CLAUDE_BIN=&lt;path-or-omit-for-PATH&gt;</code>. Per §107.x: ALL live evidence lands under <code>docs/qa/HRD-NNN-LIVE-&lt;TS&gt;/</code> with the HRD-133 sanitizer guarantee that no token text appears in any committed transcript/log. Once operator exports, conductor drives live runs + captures evidence + flips the SKIP-with-reason invariants to PASS in the next commit. <b>If operator declines / defers:</b> next priorities are (a) Wave 8 supervised channel restart (currently fail-loud-tears-down-siblings per <code>pherald/cmd/pherald/listen.go</code>: 387-398 doc-comment), (b) docs audit task #147 (comprehensive per-flavor operator guides), (c) HRD-131 §11.4.93 SQLite SSoT — remains OPEN-deferred until constitution-repo migration binary lands. <b>Prior r18 — v1.0.0 readiness (workstream 3) per plan</b> <code>docs/superpowers/plans/2026-05-27-v1.0.0-execution.md</code>; <b>Batch A + Batch B DONE, next is Batch C: (1) Batch C — §43 command catalogue HRD-029..056 (IMMEDIATE NEXT UNIT)</b> — the live CLI command bodies that PRODUCE the Subjects the Batch-B detectors classify (<code>pherald</code> commit-push, <code>sherald</code> destructive-guard, <code>cherald</code> creds-scan, <code>rherald</code> tag-mirror, <code>bherald</code> evidence-</p> |

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capture, etc.), grouped into 5 flavor clusters per the v1.0.0 plan. Closing Batch C is what flips HRD-019..025 from in\_progress→Fixed (their bindings have no live Subjects until the command surface lands). **NOTE — the e2e invariant counter is at E96; next free is E97+.** (2) **Batch D — commons\_infra HRD-081 + HRD-085..090.** (3) **Batch E — Wave 7 T6-T12 Slack adapter HRD-115..121** (vendored slack-go submodule — **verify the I6 inheritance-gate check after the .gitmodules entry**; confirm slack-go v0.16.0 field names inner.Files/socketmode.EventTypeEventsAPI/UploadFileV2Parameters; then qaherald Slack MessengerClient, spec §11.0/§32.2/§43, e2e, mutation gate, Issues→Fixed/v0.6.0/4-mirror push; T5 pre-staged perChannelConfig("slack") + blank-import in listen.go — HERALD\_CHANNELS=tgram, slack currently errors ErrUnknownChannel until T6 registers slack, by design; plan docs/superpowers/plans/2026-05-27-wave7-generic-messenger.md line 980). **PLAN GOTCHAS:** Wave 7 e2e invariants **MUST** renumber off E81-E89 — use **E90+** (E89 is now consumed by Batch A's audit write-through); the slack-go submodule addition requires re-running the I6 inheritance-gate check (tests/test\_constitution\_inheritance.sh). Tag cadence: v0.5→v1.0.0 across the batches. (5) **GAP-4 docs/qa back-fill + docs-audit #147** — features lacking docs/qa/<run-id>/ evidence per the ee63ce6 qa-coverage audit + comprehensive documentation audit. **DEFERRED / GATED ITEMS: §11.4.93 SQLite SSoT migration (HRD-131) — OPERATOR-DEFERRED 2026-05-27:** Phase 2 adoption is blocked on the constitution's workable-items migration binary, which is an unimplemented scaffold (0/7 subcommands) + schema-incompatible with Herald's HRD-NNN pipe-table format; per §11.4.74 Herald may NOT reimplement it — the fix is a constitution-repo task affecting all consumers (assessment docs/research/workable-items-phase2-assessment-2026-05-27.md, 36fa90b). HRD-131 stays OPEN, blocked-on-constitution-binary. MTPProto real-channel automation — **OPERATOR-DECISION:** provide my.telegram.org app\_id+app\_hash (folds into a qaherald MessengerClient MTPProto impl reusing the entire scenario engine; Telegram never relays bot-to-bot in groups — see docs/research/telegram-bot-to-bot-constraint.md); operator should also /revoke @pherald\_qa\_bot (leaked plaintext token). §11.4.90 obsolescence — zero current Obsolete rows; convention in place, periodic re-audit pending. (RESOLVED this session (2026-05-28 day-2): v1.0.0 Batch B COMPLETE — all 7 §42 flavor constitution bindings HRD-019..025 landed (f33b5e4/5ebad4f/46ca77e/f73195f/3fb753a), each internal/bindings/ Pipeline + rule catalogue + PURE detectors + §107.x evidence + §11.4.85 stress 1200-concurrent, e2e E90-E96 wired (HRD-019..025 stay in\_progress — live-op interception is Batch C); constitution pulled 3c9c4e9→15cd4bc + §11.4.95-97 propagated (d5b9895, gate 15/0); 3 fixes — #194 audit-test qaRoot t.TempDir guard (a8ad4f7), eventbus PublishedByType data race → LoadOrStore (f5d1367, a real E3 -race regression latent since Batch A), E17 e2e SKIP-guard now requires live PG (75e7a49). Prior r13: v1.0.0 Batch A — HRD-018/026/027 closed atomically Issues→Fixed, e2e **81 PASS / 0 FAIL** (E89 wired), commons\_constitution audit write-through §107 bluff FIXED + admin mode-flip REST; conductor-verified -race -count=3 + real-PG + §11.4.85 stress 512-row no-lost-writes. Earlier: GAP-3 §11.4.85 stress+chaos suite IMPLEMENTED — e2e 78 PASS / 0 FAIL — AND atomically migrated Issues→Fixed (HRD-122..130 Completed); HRD-132 Runner idempotency-window bug FIXED 354b883 (claim-before-dispatch, exactly-once sends==1) + migrated to Fixed.md; constitution §11.4.89-94 propagation + residue scanner; HRD-123 flaky oversized-body

| Field | Value                                                                                                                                                                                                                                                                                                                                      |
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|       | assertion + E82 stale-anchor anti-bluff issues fixed. Prior r10 RESOLVED: Wave 6.5 mutation-gate M3 stale anchor; Postgres-SASL e2e FAILs; HRD-101 filed as a real Fixed.md row.) §11.4.87 endless-loop: continue until Issues.md zero-active + CONTINUATION §3 empty + no subagent in flight. NO BLUFF. Push 4 mirrors every 2-3 commits. |

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## §0. How to use this document

Paste the following block into any CLI agent (Claude Code / OpenCode / Cursor / Aider / Gemini CLI) to resume Herald work exactly where it was left:

You are working on the Herald project at ~/Projects/Herald (also reachable as the Herald/ submodule of a consuming project). The Helix Universal Constitution lives at <ancestor>/constitution/ (parent-walk discovery). Read in this order: CLAUDE.md, AGENTS.md, README.md, docs/guides/HERALD\_CONSTITUTION.md, docs/guides/CONSTITUTION\_INHERITANCE.md, docs/specs/mvp/specification.V3.md. Then read docs/CONTINUATION.md (this file) for live state, docs/Issues.md for open work, docs/Status.md for current phase, docs/Fixed.md for closed history. Go workspace builds via go test ./commons/... ./commons\_prefix/... ./commons\_messaging/... ./commons\_storage/.... Inheritance gate tests/test\_constitution\_inheritance.sh MUST exit 0 before any commit. Multi-mirror fan-out push to four hosts (GitHub + GitLab + GitFlic + GitVerse) is mandatory per Constitution §103.

## §1. Snapshot

- **Active spec:** docs/specs/mvp/specification.V3.md Revision 4 (~4300 lines).
- **Archived specs:** V1 + V2 in docs/specs/mvp/archive/ (frozen).
- **Go modules:** commons, commons\_prefix, commons\_messaging, commons\_storage, pherald (all compile + unit tests pass).
- **Container scaffold:** quickstart/{docker-compose.quickstart.yml, Dockerfile.pherald, otel-config.yaml, .env.example} for §26.5 quickstart.
- **CLI:** pherald version --json returns canonical build info; serve/send/doctor/migrate/subscriber/deadletter stubbed with HRD-NNN pointers.
- **Inheritance gate:** 15 PASS / 0 FAIL. Meta-test ✓.
- **e2e bluff hunt:** invariants now run through **E137** (E89 Batch-A; E90-E96 Batch-B bindings; E97-E120 Batch C §43 command catalogue; E121-E131 Wave 7 multi-channel; E132-E133 HRD-137 tgram stress+chaos; E134 mtproto scaffold; **E135-E137 = Wave 8 Track B MTProto live tests — E135 Subscribe roundtrip / E136 Wave 6 closed loop /**

**E137 Wave 6.5 lifecycle**, all PASS LIVE 2026-05-28). E135-E137 honest-SKIP-with-reason when MTProto creds absent. Next free invariant id is **E138+**.

- **Stress+chaos mutation gate:** tests/test\_stress\_chaos\_mutation\_meta.sh (HRD-130) 6 PASS / 0 FAIL (5 load-bearing mutations).
- **Phase:** implementation r1.

## §2. Last commit landed

**HEAD** b313a5e — docs(qa): Wave 8 Track B live evidence — E135/E136/E137 all PASS LIVE under §11.4.98. Wave 8 Track B (MTProto user-account harness for §11.4.98 full-automation) CLOSED end-to-end: HRD-139 MTProto Client + qaherald mtproto login/whoami/logout (238448e + ec4c3c0); HRD-140 Subscribe roundtrip PASSED LIVE in 1.30s (367d58b); HRD-141 Wave 6 closed loop PASSED LIVE in 48.58s with journal-based §11.4.98 autonomy assertion (7e50ed0 + d7d6524 + 0797e85); HRD-142 Wave 6.5 lifecycle PASSED LIVE in 26.04s, 3/3 subtests (1f2e75d); E135-E137 wired into bluff hunt (a80d206); HelixConstitution §11.4.99 anchored upstream + Herald root + 9 Helix-stack submodules (fb9d81e + d1100b2 + dc007ab + d1b6f25); evidence committed under docs/qa/HRD-LIVE-MTPROTO-20260528T125321Z/ (b313a5e). v0.6.0 ANNOTATED TAG cut. **Prior** 77da6eb — feat(tgram HRD-137 + Wave C): §11.4.85 stress+chaos SKIP→PASS + 6 HRDs Issues→Fixed. **Tgram-completeness wave COMPLETE — HRD-133..138 atomic Issues→Fixed per §11.4.19.** 9 files changed (3 modified + 6 created): a one-line commons\_messaging/channels/tgram/subscribe.go seam fix (threads a.baseURL into telebot.Settings.URL before NewBot, parallel to ensureBot at tgram.go:60-67; production path unchanged when a.baseURL == ""); new commons\_messaging/channels/tgram/stress\_chaos\_test.go carrying TestTgram\_Stress\_MultiRecipientFanOut (already PASS pre-fix, Wave-B-era) + TestTgram\_Chaos\_GetUpdatesPollerResilience (httptest fault-injector — 3× 500 + 2× 1.5s hangs honoring ctx-cancel + 1× mid-body Hijacker connection-close + 2× success); docs/qa/HRD-137-20260528T150000Z/stress\_chaos/artefacts (anchor preserved verbatim, transcript §2 documents SKIP→PASS, chaos\_assertion.txt rewritten with measured PASS-state values); atomic Issues.md r34→r35 + Fixed.md r15→r16 migrating all 6 tgram-completeness HRDs (HRD-133/134/135/136/137/138) Issues→Fixed. Deterministic across -race -count=3 — every iteration identical: recovery=6 fault(500=12 hang=8 close=3 ok=6) goroutines 3→4 (leaked=1) duration=12.07s. Full tgram suite **26/26 PASS, 0 SKIP** (was 24/24 with chaos SKIPped). All 4 mirrors synced.

## §3. Active work

**No active work in flight.** Wave 8 Track B closed; v0.6.0 tag cut and pushed to all 4 mirrors. Host machine going to power-off per operator end-of-day request. Pickup tomorrow: (a) §11.4.19 atomic Issues→Fixed migration for HRD-139/140/141/142 (currently tracked only in TaskList — need Issues.md+Fixed.md rows + summary updates), (b) §11.4.90 Obsolete graduation of the 3 legacy NON-COMPLIANT tests (TestSubscribe\_LiveBotAPI, tests/test\_wave6\_live\_loop.sh, lifecycle --manual mode) with Obsolete-Details citing superseded-by-later-mandate §11.4.98 / §108.m, (c) v1.0.0 Batch C carry-over (none — all closed this session per Tasks #195-#204) so next unit is Batch E HRD-115..121 status review (already Completed per Tasks #206-#212) → confirm Issues.md / Fixed.md migration, (d) then continue with §43 §43 catalogue stragglers + GAP-4 docs/qa back-fill + comprehensive docs audit task #147. MTProto creds remain in .env (operator-supplied + session cached at ~/.config/herald/mtproto.session for next-session reuse). All hermetic gates green: tgram suite 26/26 PASS / 0 SKIP, chaos test deterministic -race -count=3. Wave 8 Track B live tests

PASSING on every rerun — fully autonomous past the single one-time OOB-code paste at MTProto login (which Telegram fundamentally requires).

## §4. Next concrete steps

1. **Live evidence — Telegram + Claude Code closed-loop** (gated on operator export). Drive `tests/test_wave6_live_loop.sh` end-to-end with real bot + chat. Capture transcript under `docs/qa/HRD-100-LIVE-20260528-<TS>/`. Flip E17/E18/E34/E63-E70 SKIP→PASS in `scripts/e2e_bluff_hunt.sh`. Per §107.x: ALL live evidence lands with the HRD-133 sanitizer guarantee (no token text in any committed transcript/log).
2. **Live evidence — Wave 6.5 ticket lifecycle** (same credentials). Drive the qaherald lifecycle scenarios against the real bot. Capture under `docs/qa/HRD-NNN-LIVE-W6.5-<TS>/`. Flip E71-E80 SKIP→PASS.
3. **Tag v0.6.0** (after E17-E80 live evidence + Slack live operator-provided OR explicit operator deferral). Wave 7 T6-T12 close at the same boundary; HRD-115..121 Issues→Fixed.
4. **Tag v0.7.0 / v1.0.0-dev-NN cadence** — operator-provided live credentials unblock the v0.6→v1.0 chain (mirrors the v0.4.0 / v0.5.0 deferred-until-live-evidence pattern).
5. **Deferred / gated items:** §11.4.93 SQLite SSot (HRD-131, constitution-repo task), MTProto real-channel automation (operator decision), Wave 8 supervised channel restart (currently fail-loud-tears-down-siblings per `pherald/cmd/pherald/listen.go:387-398` doc-comment), docs audit task #147 (comprehensive per-flavor operator guides).

### §4a. Wave 6 live-test handoff (T10b — gating tag v0.4.0)

**Operator-supplied credentials required** — Wave 6's e2e invariants E63-E70 land as honest SKIPs until the live closed-loop runs against real Telegram + real Claude Code with real chat messages exchanged.

#### Step-by-step.

1. Export credentials in the shell that will drive the test:

```
export HERALD_TGRAM_BOT_TOKEN="<your-bot-token-from-BotFather>"
export HERALD_TGRAM_CHAT_ID="<numeric-chat-id-where-bot-is-admin>"
export HERALD_CLAUDE_BIN="$(command -v claude)"      # path to the claude CLI binary
export HERALD_PROJECT_NAME="Herald"                  # or any operator-chosen session name
```

2. Pick a `<run-id>` (timestamp + 4-char nonce):

```
RUN_ID="$(date -u +%Y-%m-%dT%H-%M-%S)-$(uuidgen | cut -c1-4)"
QA_DIR="docs/qa/HRD-100-{$RUN_ID}"
mkdir -p "{$QA_DIR}/attachments"
```

3. Type ONE message in the configured Telegram chat (script reads it via `getUpdates` within 60s).

#### 4. Run the live closed-loop test:

```
bash tests/test_wave6_live_loop.sh
```

The script builds pherald, observes the original message\_id, starts pherald listen --bot-token "\$HERALD\_TGRAM\_BOT\_TOKEN" --chat-id "\$HERALD\_TGRAM\_CHAT\_ID" --qa-out-dir "\$QA\_DIR" in the background, waits up to 45s for a reply with reply\_to\_message\_id == original, and exits 0 on PASS.

#### 5. Capture supplementary logs:

```
cp /tmp/pherald-w6.log "${QA_DIR}/pherald-listen.log"
# claude stdout/stderr — paths depend on the journaling
# setup. Copy whatever
# the test produced into ${QA_DIR}/claude-stdout.log +
# claude-stderr.log.
```

6. Author a brief \${QA\_DIR}/README.md (5 lines minimum) narrating: who ran it, when, what message was sent, what reply came back, any anomalies. This is NOT auto-generated — operator-written narrative is the §107.x evidence anchor.
7. git add \${QA\_DIR}/ && git commit -m "Wave 6 step 10b: docs/qa/HRD-100-\${RUN\_ID}/ live closed-loop evidence" then proceed to T13b (v0.4.0 tag + 4-mirror push).

**If the script SKIPS** (any env var unset): the test prints SKIP: <reason> and exits 0. That is honest §11.4.3 hardware-absent SKIP-with-reason, not a PASS. Tag v0.4.0 MUST NOT be created until at least one PASS run is committed to docs/qa/HRD-100-<run-id>/.

## §4b. qaherald-auto resume runbook (next session — gating tag v0.5.1)

#### Status as of 2026-05-23 session end:

- HEAD 8ad4cb9 on main; all 4 mirrors in sync (github.com:vasic-digital/Herald.git, gitlab.com:vasic-digital/herald.git, gitflic.ru:vasic-digital/herald.git, gitverse.ru:vasic-digital/Herald.git).
- Tag v0.4.0 shipped (Wave 6 closed-loop). Tag v0.5.0 shipped (Wave 6.5 lifecycle implementations — premature: only S1+S2 live evidence + carries the pre-53a7ad3 envelope-action-guidance bug). Tag v0.5.1 is the target after qaherald-auto T6-T8 land + full 15-scenario evidence committed.
- Plan: docs/superpowers/plans/2026-05-23-qaherald-auto.md (commit 52ce679).
- Done: T1 skeleton + T2 messenger + T3 scenarios + T4 orchestrator + T5 report (combined commit 8ad4cb9 for T3-T5).
- TODO: **T6 pre-flight** + **T7 hermetic tests** + **T8 shell-adapter** then live run + tag.

### Resume prerequisites

The following are **already provisioned** in ~/ .zshrc (export-on-shell-start):



```

HERALD_TGRAM_BOT_TOKEN
    # @atmosphere_worker_bot — pherald-bot (Telegram bot
    8823384001)
HERALD_TGRAM_CHAT_ID      # -4946584787 (ATMOSphere Development
    group)
HERALD_OPERATOR_IDS       # 2057253161 (the operator's user-id)
HERALD_QA_BOT_TOKEN       # @pherald_qa_bot — 2nd bot
    (8971749017, privacy DISABLED, member of the group)

```

**The qa-bot token is plaintext in** `~/ .zshrc` — operator should `/revoke` via `@BotFather` after Wave 6.5 close-out and regenerate (the token leaked into one prior session's bash output earlier today, was never committed to git but lives in `/private/tmp/claude-501/...` operator-local files).

The containers `herald-postgres` (port 24100) and `herald-redis` (port 24200) are LIVE — both have been Up since the start of this session; data persists across podman restarts. Schema is at migration v12 (all 12 applied).

## Resume sequence (suggested next-session order)

### 1. Pull state + verify:

```

cd ~/Projects/Herald && git pull origin main && git log --
    oneline -10
git status --short # expect clean
git stash list     # 2 stashes preserved: T8 Wave-5 salvage
    + premature S3-S5 transcript WIP

```

2. **Spawn T6 (pre-flight)** via a fresh subagent — read `docs/superpowers/plans/2026-05-23-qaherald-auto.md` Task 6 in full. Implementation lands at `qaherald/internal/lifecycle/preflight.go` + `preflight_test.go`. 10 gates: `pherald-bot` reachable via `getMe`, `qa-bot` reachable, `qa-bot` privacy disabled, `qa-bot` is group member, `OPERATOR_IDS` contains `qa-bot`, etc. Each gate has distinct exit code for diagnostics.

3. **Spawn T7 (full hermetic tests)** — extends T3-T5's 12 tests with `httptest`-based scenario simulation. Coverage targets: every scenario PASS path + every FAIL diagnostic + S9 SKIP path + S14 outbound-attachment sha256 round-trip + S11/S12/S13 inbound-attachment download + `Issues.md/Fixed.md` fs-mutation assertion.

4. **Spawn T8 (shell-script adapter)** — rewrites `tests/test_wave6.5_lifecycle.sh` to delegate to `qaherald lifecycle` for automated runs. Keeps the original `--manual` flag for the operator-typing interactive UX as a fallback.

### 5. Run the live e2e (the §107 watershed):

```

cd ~/Projects/Herald
# Add qa-bot user-id to OPERATOR_IDS so S5/S6/S8/S10 succeed
export HERALD_OPERATOR_IDS="$
    {HERALD_OPERATOR_IDS},8971749017"
# Re-export from .zshrc (or just open a new shell)

```



```

source ~/.zshrc

# Build pherald + qaherald
go build -o /tmp/pherald ./pherald/cmd/pherald
go build -o /tmp/qaherald ./qaherald/cmd/qaherald

# Start pherald listen in background with QA journaling
RUN_ID="$(date -u +%Y-%m-%dT%H-%M-%S)-w6.5live"
PHERALD_QA_DIR="docs/qa/HRD-101-lifecycle-`${RUN_ID}`-pherald"
QAUTO_QA_DIR="docs/qa/HRD-101-lifecycle-`${RUN_ID}`"
mkdir -p "${PHERALD_QA_DIR}/attachments" "${QAUTO_QA_DIR}/
    attachments"
/tmp/pherald listen --qa-out-dir "${PHERALD_QA_DIR}" --docs-
    dir docs &
PHERALD_PID=$!
trap 'kill -TERM $PHERALD_PID 2>/dev/null' EXIT
sleep 5

# Run qaherald lifecycle (T6-T8 must be done first)
/tmp/qaherald lifecycle \
    --pherald-bot-username=atmosphere_worker_bot \
    --pherald-qa-out-dir="${PHERALD_QA_DIR}" \
    --out="${QAUTO_QA_DIR}" \
    --run-id="${RUN_ID}"
# Exit 0 on all-PASS or all-PASS-with-S9-SKIP. Non-zero on
    any FAIL.

```

## 6. Capture evidence:

- `\${PHERALD\_QA\_DIR}`/transcript.jsonl — pherald's view (classifications, dispatch, replies)
- `\${QAUTO\_QA\_DIR}`/transcript.jsonl — qaherald-auto's view (sends, assertions)
- `\${QAUTO\_QA\_DIR}`/report.md — Markdown summary
- `\${QAUTO\_QA\_DIR}`/attachments/<sha256>.<ext> — content-addressed inbound + outbound attachments
- docs/Issues.md + docs/Fixed.md — mutated by S5/S6/S8/S10 (and reverted by S10 + S15 cleanups; both files end in valid state)

## 7. Author operator README + commit:

```

cat > "${QAUTO_QA_DIR}/README.md" <<EOF
# HRD-101 Wave 6.5 lifecycle evidence — <run-id>

<5+ lines narrating: who ran it, when, which scenarios
    PASSEd/FAILEd/SKIPped,
bot reply quality observations, any anomalies. Operator-
    written, NOT auto-generated.>
EOF

```

```
git add docs/qa/HRD-101-lifecycle- $\{\text{RUN\_ID}\}$ *
git commit -m "Wave 6.5 step 13: HRD-101 live 15-scenario
lifecycle evidence (qaherald-auto)"
```

#### 8. Tag v0.5.1 + 4-mirror push:

```
git tag -a v0.5.1 -m "Wave 6.5 close: full lifecycle live
evidence + envelope action-guidance fix"
git push origin main
git push origin v0.5.1
# Verify all 4 mirrors converged on the same SHA + tag
for mirror in github.com:vasic-digital/Herald.git
gitlab.com:vasic-digital/herald.git
gitflic.ru:vasic-digital/herald.git
gitverse.ru:vasic-digital/Herald.git; do
git ls-remote "git@ $\{\text{mirror}\}$ " v0.5.1
done
```

### Open / known constraints for tomorrow

- **S9 (non-operator Done: rejection):** requires `HERALD_QA_BOT_TOKEN_NON_OPERATOR` env (a THIRD bot account whose user-id is NOT in `OPERATOR_IDS`). Without it, S9 emits a SKIP-with-reason. Operator may register a third bot OR accept the SKIP (S9's logic is already unit-tested hermetically in T5's mutation gate).
- **First-scenario CC bootstrap:** ~30s on first inbound message because the Claude Code session must spawn. Subsequent scenarios use the cached session (~5-15s typical).
- **Issues.md mutation race:** if you run `qaherald lifecycle` while ANOTHER pherald listen is also processing in the same checkout, Issues.md may receive double-writes. Single-pherald-listen invocation is the safe default.
- **Postgres-SASL carry-over (RESOLVED 2026-05-27):** the `e2e_bluff_hunt` FAILs on Postgres-SASL invariants (E7-E12 + E14-E16 + E37-E42) are FIXED — `9b166c4` (auth env + PG password self-heal + Redis/tenant wiring + honest SKIP-with-reason) + `bdb9f1` (Runner nil-Redis graceful PG-only fallback). At resolution `e2e_bluff_hunt` reported 60 PASS / 0 FAIL / 24 SKIP; the later GAP-3 session (`d2a60d3`) raised it to **78 PASS / 0 FAIL** with E81-E88 wired in.

### Safe-shutdown checklist (HISTORICAL — end of 2026-05-23 session; superseded — current HEAD is `d2a60d3`)

| Check                                  | State                                                                                                                                                |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| All committed work pushed to 4 mirrors | YES — HEAD <code>8ad4cb9</code> confirmed on github+gitlab+gitflic+gitverse                                                                          |
| Local working tree clean               | YES — <code>git status --short</code> empty                                                                                                          |
| Stashes preserved                      | 2 stashes: T8 Wave-5 salvage (older) + premature mid-test S3-S5 transcript WIP (newer, will be regenerated cleanly by qaherald-auto in next session) |
| Background pherald/qaherald processes  | NONE running ( <code>ps aux   grep pherald</code> empty)                                                                                             |

| Check            | State                                                                                                                                                                                                                                      |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Container state  | herald-postgres + herald-redis Up 17 hours — persist data across podman stop && podman start; can be left running OR stopped (podman stop herald-postgres herald-redis); next session restart is podman start herald-postgres herald-redis |
| /tmp/*.log files | one operator-token leak shred-deleted earlier today; remaining transient logs contain no credentials (verified)                                                                                                                            |
| .zshrc           | contains 4 HERALD_* exports including HERALD_QA_BOT_TOKEN plaintext — operator should /revoke + regenerate via @BotFather post-Wave-6.5 closure                                                                                            |
| Memory entries   | All session findings persisted under /Users/milosvasic/.claude/projects/-Users-milosvasic-Projects-Herald/memory/ — survive across sessions                                                                                                |

**Machine is safe to shut down.** Resume tomorrow by `cd ~/Projects/Herald && git pull origin main` and start with qaherald-auto T6.

## §5. Long-form pointers

- docs/specs/mvp/specification.V3.md — full active spec (Revision 4).
- docs/specs/mvp/specification.V3.md#30-v2-self-review-log — every review pass.
- docs/guides/HERALD\_CONSTITUTION.md — §101..§106 extending Universal.
- docs/guides/CONSTITUTION\_INHERITANCE.md — parent-discovery + gate.
- tests/test\_constitution\_inheritance.sh — the gate.
- quickstart/ — HRD-008 scaffold.
- commons/types.go — the §11.0 type contract reference.